

AllelePerturb reveals measurement limits of protein-coding variant prediction in single-cell transcriptomics

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Abstract

Single-cell perturbation prediction is becoming a central task for virtual-cell modeling, but current methods and benchmarks largely define perturbations at gene, drug or condition resolution. This design is suited to knockout and drug-response screens, yet does not address how different protein-coding variants within the same gene alter cellular state. Here we introduce AllelePerturb, a benchmark and evaluation framework for held-out protein-coding variant prediction in single-cell transcriptomics. AllelePerturb integrates 472 variants across TP53, KRAS, GATA1 and JAK1, comprising 321,043 single cells from Perturb-seq, base editing and scSNV-seq, and evaluates prediction quality across direction recovery, allele-level discrimination and differential-expression fidelity.

Across 20 feature-model combinations, all predictors recover perturbation direction (Pearson- δ 0.59–0.67), yet none reliably distinguishes held-out variants (PDS 0.43–0.49, at or below chance). Split-half analysis traces this direction-ranking dissociation to a measurement-resolution floor governed by variant-to-wild-type effect size, within-variant sampling noise and effective cells per variant. The same floor appears in gene-level Perturb-seq atlases, where roughly half of Replogle perturbations are un-rankable at native depth. AllelePerturb reframes variant-level perturbation prediction as a power-aware evaluation problem and provides effect-size-conditioned guidance for future variant-level screens.

Introduction

Single-cell perturbation prediction is becoming a central testbed for virtual-cell modeling. High-throughput perturbation screens now make it possible to measure how cell populations respond to genetic and chemical interventions at scale, and computational models are increasingly expected to predict the transcriptional consequences of perturbations that have not been experimentally observed. A growing ecosystem of generative, optimal-transport, graph-based and foundation-model approaches has therefore been developed for this task^{1–5}, and community benchmarks have begun to standardize their comparison across datasets, cell types and perturbation classes^{6,7}. These efforts have established perturbation prediction as a useful abstraction for drug screening, functional genomics and cellular response modeling.

Yet the field still largely operates at the resolution of genes, drugs or experimental conditions. In most current benchmarks, a perturbation is represented as a gene identity, drug identity or dosage condition: the model is asked to predict the effect of knocking down TP53, perturbing

KRAS or applying a compound. This design is well matched to gene-knockout and drug-response screens, but it leaves unresolved a finer and clinically central question: how do different protein-coding variants within the same gene alter cellular state? This distinction matters because many disease mechanisms are allele-specific. TP53, for example, carries hundreds of missense mutations with divergent consequences: R175H disrupts protein folding, R273C preserves the folded structure but impairs DNA binding, and R248Q has been associated with dominant-negative and gain-of-function behavior⁸. Similarly, KRAS G12D and G12V are both activating codon-12 mutations, but they differ in downstream signaling behavior and biological effect size⁹. A model that predicts the average consequence of a TP53 perturbation is therefore not necessarily a model that distinguishes TP53 R175H from TP53 R273C. Variant-effect prediction has been extensively benchmarked using multiplexed assays of variant effect and deep mutational scanning, which quantify variant fitness or molecular function at scale^{10,11}. However, these resources provide scalar functional scores rather than population-level single-cell transcriptional responses.

Variant-level single-cell measurements and variant-conditioned models have begun to appear, but they do not yet establish a general framework for evaluating held-out protein-coding variant prediction. Single-cell expression-based variant-impact phenotyping has shown that TP53 and KRAS variants can produce reproducible expression signatures when assayed at sufficient depth¹², and PerturbNet introduced variant-level perturbations into conditional generative modeling in a single-gene GATA1 setting¹³. Recent benchmarking studies have also shown that perturbation-prediction metrics and model rankings can be sensitive to simple baselines, systematic variation and metric geometry^{7,14–16}. However, these efforts primarily address gene- or drug-level prediction, or variant phenotyping within measured perturbations. They do not answer whether allele-specific responses within the same gene are distinguishable, predictable or even evaluable under realistic single-cell sampling depth. This evaluation problem is fundamental: standard metrics such as mean squared error or Pearson correlation can reward recovery of the shared direction of a gene-level response even when a model fails to discriminate individual alleles. If split-half variability within a variant approaches the variant-to-wild-type signal, then the measurement itself leaves little signal window for allele-specific ranking.

Here we introduce AllelePerturb, a benchmark and evaluation framework designed specifically for held-out protein-coding variant prediction in single-cell transcriptomics. AllelePerturb-Bench integrates 472 protein-coding variants across four genes (TP53, KRAS, GATA1 and JAK1) and 321,043 single cells spanning three experimental technologies: Perturb-seq, base editing and scSNV-seq. AllelePerturb-Eval decomposes prediction quality into three complementary dimensions: direction recovery, allele-level discrimination and differential-expression fidelity. It further defines six generalization splits that test random generalization, positional extrapolation, mechanistic extrapolation, cross-gene transfer, low-depth evaluation and compatibility with existing protocols.

Using this framework, we systematically evaluated 20 feature-model combinations and found a consistent direction-ranking dissociation. All predictors recovered the transcriptional direction of perturbation effects, with Pearson- δ values between 0.59 and 0.67, yet none reliably distinguished the responses of held-out variants, with perturbation discrimination scores between 0.43 and 0.49, at or below the chance expectation of 0.50. Split-half analysis traced this dissociation to a measurement-resolution floor governed by the ratio between variant-to-wild-type effect size, within-variant sampling noise and effective cells per variant. Extending the same analysis to widely used gene-level Perturb-seq atlases showed that this floor is not unique to allele-level data: at native depth, roughly half of perturbations in the Replogle atlas were un-rankable by direct split-half energy distance. AllelePerturb therefore reframes protein-coding variant prediction from a simple model-ranking problem into a power-aware evaluation problem, and provides

a standard framework for determining when variant-level single-cell perturbation prediction is measurable enough to benchmark.

Results

AllelePerturb defines perturbation prediction at protein-coding variant resolution

To evaluate perturbation prediction at protein-coding variant resolution, we defined the prediction target as a held-out protein-coding variant rather than a held-out gene, drug or condition. In this setting, the model must predict the transcriptional effect of an unseen allele from its variant features. This differs from gene-level perturbation prediction, in which all perturbations of the same gene are treated as a single condition. In TP53, for example, a gene-level representation collapses variant responses into a common perturbation direction (Fig. 1a, top), whereas allele-resolution analysis preserves variant-specific departures from wild type, with mutations such as R175H, R273C and R248Q occupying distinct regions of the response space (Fig. 1a, bottom). AllelePerturb therefore treats within-gene variant identity as the object of prediction rather than as experimental annotation.

AllelePerturb-Bench integrates four variant-perturbation datasets spanning different proteins, cell types and experimental technologies (Fig. 1b,c). TP53 (393 aa, 98 variants) and KRAS (189 aa, 93 variants) were profiled by Perturb-seq in A549 cells¹², GATA1 (413 aa, 255 variants) by base editing in hematopoietic stem and progenitor cells¹³, and JAK1 (1,154 aa, 26 variants) by scSNV-seq in HT-29 cells¹⁷. Together, these datasets comprise 472 protein-coding variants and 321,043 single cells. Per-variant cell depth varies substantially across genes, with median depths of 929 and 658 cells per variant for TP53 and KRAS, 355 for GATA1 and 104 for JAK1 (Fig. 1c), providing a natural range of sampling regimes for testing variant-level evaluation.

To make held-out variant prediction possible, each variant was represented by protein-level features rather than by a variant identifier. We encoded variants using a 6-dimensional biophysical vector θ , including changes in hydrophobicity, side-chain volume and charge, together with fold-core location, functional-switch residue status and hotspot/pathogenic annotation (Fig. 1d). These features describe variant properties beyond gene identity and allow variants from different genes to be placed in a shared feature space. AllelePerturb-Eval then separates prediction quality into three dimensions: direction recovery, allele-level discrimination and differential-expression fidelity. Pearson- δ measures agreement between predicted and observed perturbation directions, whereas PDS directly tests the defining allele-level requirement: whether a predicted response identifies its target variant among other held-out variants (Fig. 1e). Six split strategies further test random generalization, positional extrapolation, mechanistic extrapolation, cross-gene transfer, low-depth evaluation and compatibility with existing protocols. This separation of direction recovery from allele discrimination sets up the central question of the benchmark: whether models can distinguish alleles, rather than merely recover the shared direction of a perturbed gene.

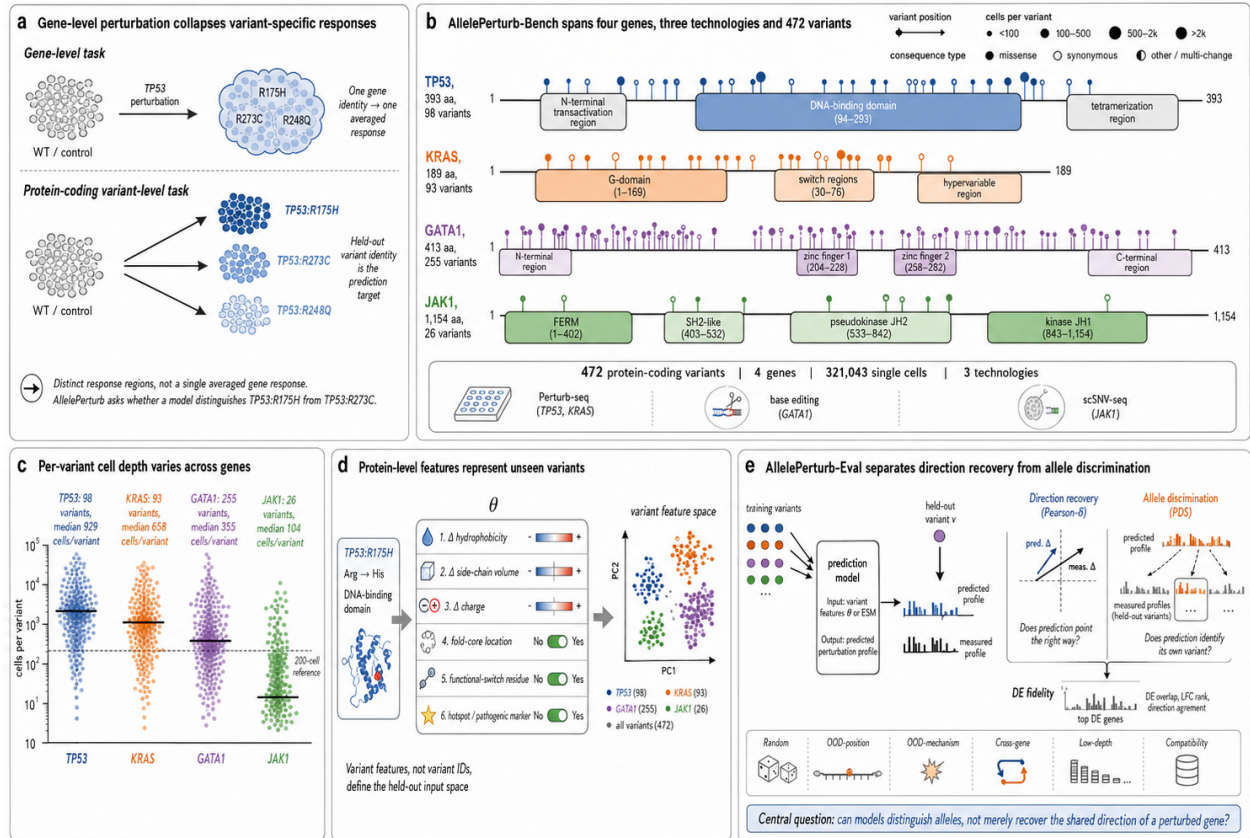


Figure 1: AllelePerturb defines perturbation prediction at protein-coding variant resolution.

a, Conceptual distinction between gene-level and protein-coding variant-level perturbation prediction. Gene-level perturbation identity collapses responses from different variants of the same gene into a single averaged effect, whereas AllelePerturb treats a held-out protein-coding variant as the prediction target and asks whether models can distinguish allele-specific responses within the same gene. **b**, Overview of AllelePerturb-Bench. The benchmark integrates 472 protein-coding variants across four genes and protein architectures (TP53, KRAS, GATA1 and JAK1), spanning three experimental technologies: Perturb-seq, base editing and scSNV-seq. **c**, Per-variant cell depth varies substantially across genes, providing a range of sampling regimes for evaluation (98, 93, 255 and 26 variants; 83.4k, 83.6k, 149.2k and 4.9k cells for TP53, KRAS, GATA1 and JAK1, respectively). **d**, Each variant is represented by a 6-dimensional biophysical vector θ encoding changes in hydrophobicity, side-chain volume and charge, together with fold-core location, functional-switch residue status and hotspot/pathogenic annotation. **e**, AllelePerturb-Eval separates prediction quality into direction recovery (Pearson- δ), allele-level discrimination (PDS) and differential-expression fidelity, and defines six split strategies to test random generalization, positional extrapolation, mechanistic extrapolation, cross-gene transfer, low-depth evaluation and compatibility with existing protocols.

Variant-level prediction exposes a direction-ranking dissociation

Having defined allele-level discrimination as the central requirement of the task, we next asked whether predictors that recover the direction of a perturbation also identify the correct held-out variant. We evaluated 20 predictors or references, comprising 18 feature-model combinations (6 model heads by 3 feature spaces), the Gene-mean predictor and the WT-null reference (Fig. 2).

Across the 19 non-null predictors, allele discrimination remained at chance: PDS-cosine ranged from 0.43 to 0.49, and no predictor exceeded the 0.50 chance level by 95% bootstrap confidence interval (Fig. 2a, left). Sixteen of 19 predictors had intervals entirely below chance and the remaining three overlapped chance. This conclusion did not depend on the analytic 0.50 baseline: within-split label permutation gave an empirical null mean of 0.500 (range 0.495 to 0.504 across 340 method-split-gene combinations), and only 3.2% of combinations exceeded the permutation 95th percentile, below the 5% false-positive rate expected by chance. PDS-L1 and PDS-L2 gave the same conclusion.

In contrast, the same predictors consistently recovered the transcriptional direction of perturbation effects. Pearson- δ ranged from 0.59 to 0.67 across predictors, and every non-null predictor had a 95% confidence interval entirely above zero (Fig. 2a, right). Plotting PDS against Pearson- δ placed all predictors in the same regime: direction recovered, allele ranking failed (Fig. 2b). Method-to-method variation was small compared with the separation between these two evaluation axes, indicating that variant-level prediction exposes a systematic direction-ranking dissociation rather than a failure of a particular model class.

Differential-expression fidelity provided a biological interpretation of the same dissociation (Fig. 2c). Direction agreement reached 77%, whereas DE-LFC Spearman was 0.45 and DE overlap was only 28%. This gradient (direction 77%, LFC ranking 45%, gene-set overlap 28%) suggests that predictors learn a program-level perturbation direction but progressively lose the finer signal needed to rank effect sizes and identify variant-specific response genes.

The magnitude of the dissociation varied by gene (Fig. 2d). It was strongest for TP53 (delta = 0.31), intermediate for KRAS (0.18) and GATA1 (0.15), and weakest for JAK1 (0.02). This ordering anticipated the split-half signal-to-noise analysis below: JAK1 had the strongest per-variant signal and the weakest dissociation, whereas TP53 and KRAS lay near the noise floor. At the single-variant scale, the same pattern appeared within TP53: Ridge-esm produced stable Pearson- δ values across representative held-out variants (0.74 to 0.82), but PDS varied from 0.00 to 0.88, with 4 of 8 examples below chance (Fig. 2e). These results suggest that allele-ranking failure is not explained by model choice alone and motivate a direct test of whether the ground truth itself is measurable at variant resolution.

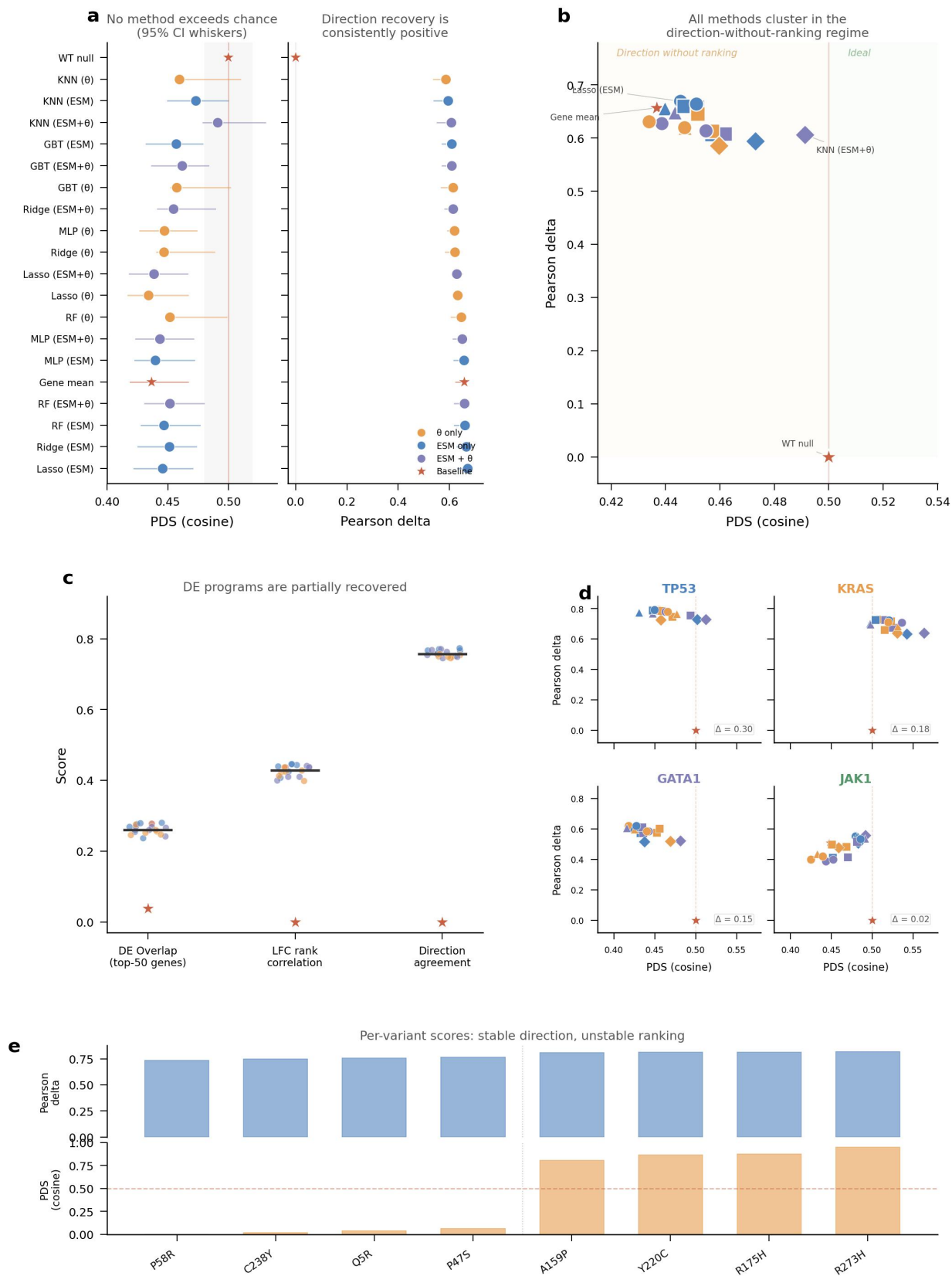


Figure 2: All methods recover perturbation direction but cannot rank variant identity. **a**, PDS (cosine) and Pearson- δ for 20 feature-model combinations with 95% bootstrap confidence whiskers; no method exceeds the chance PDS of 0.5. **b**, Methods cluster in the direction-without-ranking regime. **c**, Differential-expression fidelity: direction agreement 78%, DE-LFC Spearman 0.46, DE overlap 28%. **d**, The dissociation gap follows a per-gene gradient (TP53 0.30, KRAS 0.18, GATA1 0.15, JAK1 0.02). **e**, Per-variant scores show stable direction but unstable ranking.

Allele discrimination is constrained by an effect-size-to-noise window

The direction-ranking dissociation raised a mechanistic question: do predictors fail to rank alleles because the models are wrong, or because the ground-truth variant responses are not stably measurable at the available depth? To test the measurement component directly, we performed a split-half reproducibility analysis. For each variant, cells were randomly divided into two halves, and we compared the energy distance between the two halves of the same variant (D_{self}) with the distance between that variant and wild type (D_{null} ; Fig. 3a). A $D_{\text{self}}/D_{\text{null}}$ ratio near 1 indicates that within-variant sampling variation is comparable to the variant-to-wild-type signal, leaving little measurement window for allele-specific ranking. A ratio well below 1 indicates that the perturbation signal exceeds the split-half noise and should be rankable by distributional metrics.

The four genes occupied distinct regimes (Fig. 3b). TP53 ($D_{\text{self}}/D_{\text{null}} = 0.96$, 95% CI 0.95–0.98), KRAS (1.00, 0.99–1.02) and GATA1 (0.88, 0.86–0.90) lay near the measurement floor, whereas JAK1 was well separated from it (0.21, 0.12–0.33). This ordering matched the dissociation pattern in Fig. 2d: the genes closest to the floor showed the strongest separation between direction recovery and allele discrimination, whereas JAK1 served as a positive control with weak dissociation. GATA1 occupied an intermediate regime, with a long tail of individually rankable variants. These results show that the relevant quantity is not raw cell depth alone but the effect-size-to-noise window available for discrimination.

This point was clearest when effect size and depth were considered together (Fig. 3c,d). JAK1 had the lowest nominal depth among the four genes (median 104 cells per variant) but the widest effect-size-to-noise window, and it was already saturated in the split-half detection assay at 50 cells per half. In contrast, TP53 and KRAS had much higher nominal depths but narrow allele-specific effect windows, remaining close to the noise floor even with median raw depths of 929 and 1,000 cells per variant. Subsampling over 50 to 300 cells per half confirmed the same ordering: JAK1 remained fully detectable across the range, whereas TP53, KRAS and GATA1 increased only modestly and plateaued below 75%. Within the observed depth range, adding cells helps only when the underlying effect window is wide enough for additional sampling to resolve it.

We next asked whether this measurement floor was specific to allele-level perturbation data. Using the same native-depth, all-perturbation, direct split-half energy-distance definition, we applied the rankability analysis to three widely used gene-level Perturb-seq atlases (Fig. 3e). The three low-effect allele-level genes (TP53, KRAS and GATA1) were 98–100% un-rankable by this definition, whereas JAK1 remained a positive control with only 10% un-rankable variants. Among gene-level atlases, Replogle 2022 showed 55.3% un-rankable perturbations (95% CI 52.8–57.5%), compared with 14.6% for Adamson 2016 and 3.4% for Norman 2019. Results were qualitatively robust under alternative rankability criteria, including bootstrap-CI-based, increased-stringency and ratio-threshold definitions (Extended Data Fig. 1). Thus, the detection floor is not a peculiarity of allele-level data: finite-depth distributional distances can leave a substantial fraction of perturbations below the resolution required for stable method ranking.

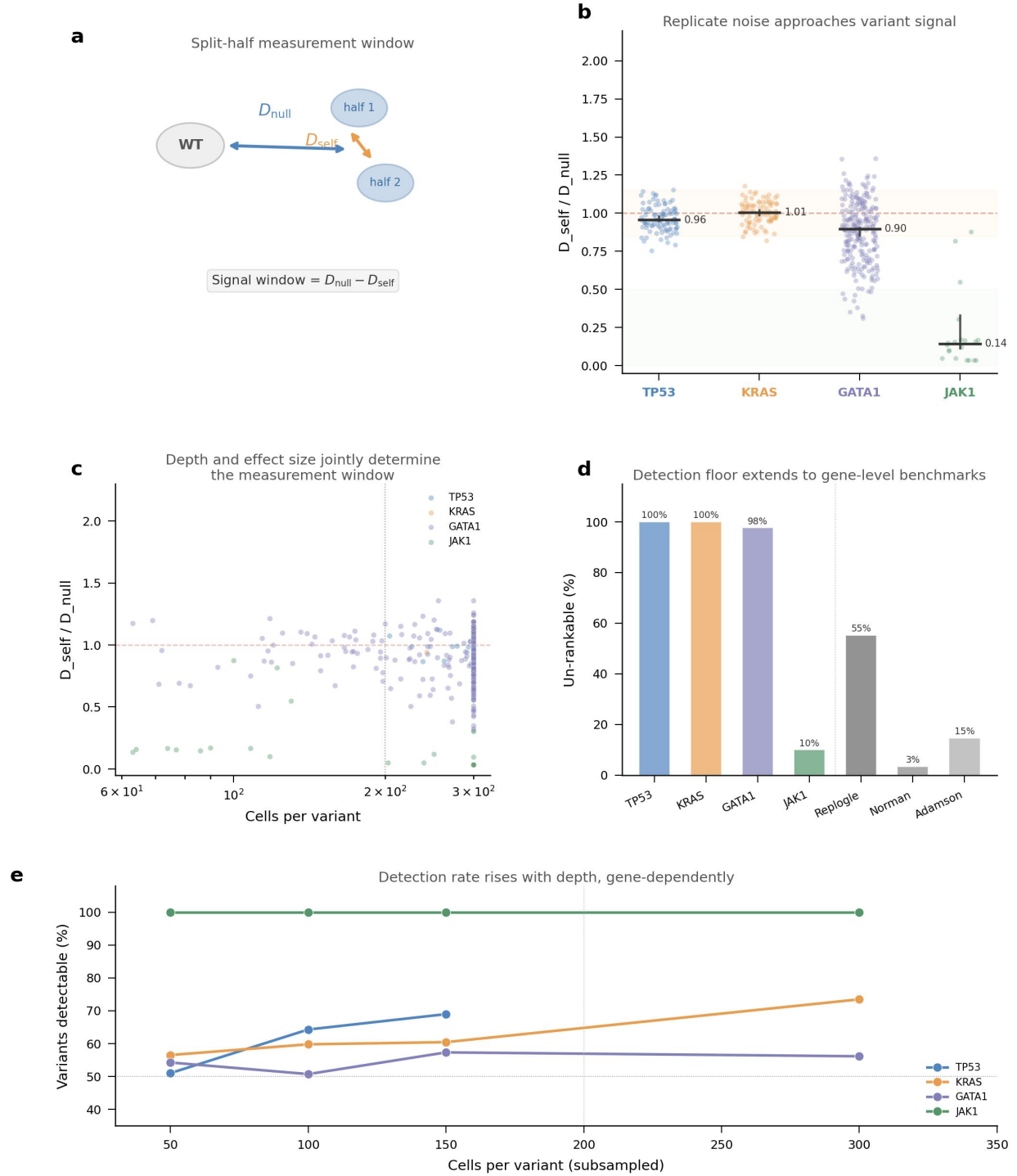


Figure 3: Split-half analysis reveals a narrow allele-level measurement window. **a**, Definition of the measurement window from within-variant (D_{self}) and variant-to-WT (D_{null}) distances. **b**, Replicate noise approaches the variant signal: $D_{\text{self}}/D_{\text{null}} = 0.96, 1.01, 0.90$ and 0.14 for TP53, KRAS, GATA1 and JAK1. **c**, Depth and effect size jointly determine the window. **d**, The detection floor extends to gene-level benchmarks (Replogle 55%, Norman 3%, Adamson 15% un-rankable), establishing that the floor is not specific to allele-level tasks (diagnostic; compare Fig. 5d). **e**, Detection rate rises with depth in a gene-dependent manner.

Allele-ranking failure persists across evaluation choices and model interfaces

We next tested whether the allele-ranking bottleneck could be explained by evaluation design or variant representation rather than by the measurement regime identified above. Across five generalization splits, Pearson- δ and PDS behaved differently (Fig. 4a,b). Pearson- δ remained positive under random, positional-extrapolation and compatibility splits, typically between 0.55 and 0.75, and declined mainly under mechanistic extrapolation and low-depth evaluation. By contrast, PDS remained near the empirical chance regime across all splits, with the strongest degradation under the low-depth split. No split produced a consistent above-chance allele-discrimination regime, indicating that the direction-ranking dissociation was not specific to the random holdout setting.

The conclusion was also insensitive to the distance function used for PDS (Fig. 4c). Cosine, L1 and L2 distances yielded nearly identical PDS distributions across the 18 feature-model combinations, with medians of 0.464, 0.476 and 0.474, respectively. Shifting from methods to gene-by-split combinations gave the same picture (Fig. 4d): Pearson- δ exceeded PDS in 15 of 17 combinations, with the two exceptions occurring in small held-out mechanistic-extrapolation sets where both metrics were close. The GATA1 low-depth split had a PDS of 0.19, consistent with the detection-limit prediction from the split-half analysis.

We then asked whether the bottleneck reflected insufficient variant representation. Increasing the input representation from the 6-dimensional biophysical θ vector to 1,280-dimensional ESM-1v embeddings did not rescue allele ranking (Fig. 4e). ESM-1v matched θ on direction recovery (Pearson- δ 0.64 versus 0.62), and concatenating ESM with θ added no gain (0.63). Yet all three feature spaces had the same PDS of approximately 0.45. Thus, within the tested representation classes, direction recovery was already saturated and the ranking bottleneck was not explained by feature dimensionality.

Finally, we audited whether existing single-cell perturbation prediction tools expose interfaces suitable for held-out protein-coding variant prediction (Fig. 4f). We treated this as an interface-compatibility audit rather than a definitive comparison of model families. The public CPA dosage interface accepts scalar dosage inputs but does not expose a channel for a multi-dimensional variant feature vector; public CellFlow and scDFM interfaces provide no prediction path for variants outside the training set; the public GEARS implementation represents genes as graph nodes and is therefore allele-blind without architectural modification; and STATE required per-gene retraining and exceeded memory at GATA1 scale in our setup. Biolord, the one external method we ran to completion, accepted θ as ordered attributes but still scored near chance (PDS 0.47–0.51). These results show that current public model interfaces were largely designed around gene- or drug-level perturbation identity and are not yet allele-resolution-ready without nontrivial modification.

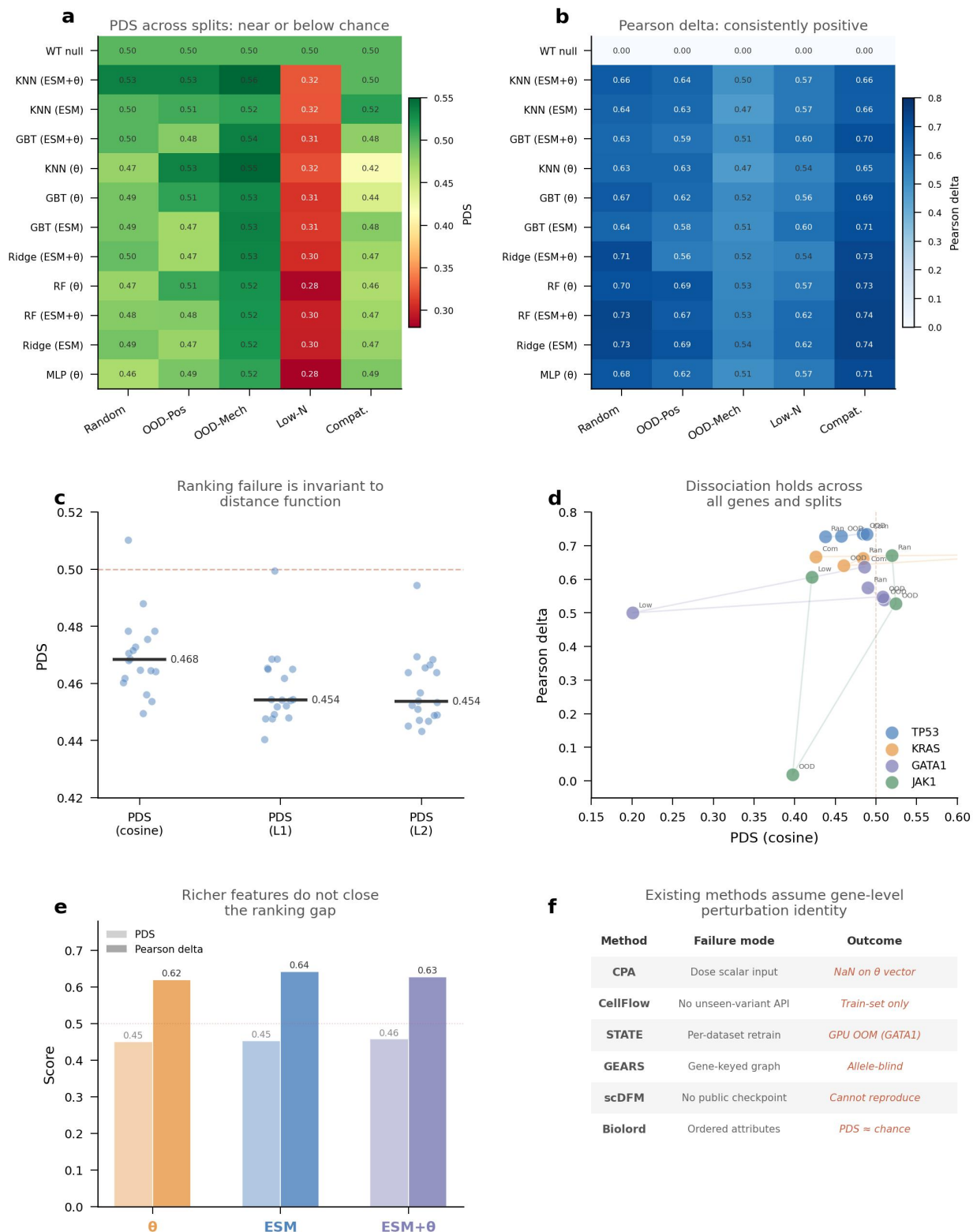


Figure 4: Allele-ranking failure persists across evaluation choices and model interfaces. **a**, PDS across five generalization splits remains at or below chance. **b**, Pearson- δ stays consistently positive across splits. **c**, Ranking failure is invariant to the distance function (PDS cosine 0.468, L1 0.454, L2 0.454). **d**, The dissociation holds across all genes and splits. **e**, Richer features do not close the ranking gap (θ 0.45, ESM 0.45, ESM+ θ 0.46). **f**, Existing methods assume gene-level perturbation identity and fail on allele-resolution input.

Rankability prediction enables power-aware evaluation of variant perturbation screens

If allele discrimination is meaningful only above a measurement floor, then variant-level benchmarks need a way to identify which perturbations are evaluable before full model comparison. We therefore asked whether rankability could be predicted from pre-modeling, pilot-estimable quantities. Using leave-one-dataset-out (LODO) cross-validation across the four AllelePerturb datasets and three external Perturb-seq atlases, we trained a logistic-regression rankability predictor from features including split-half effect size, cells per perturbation and gene-space dimensionality (Fig. 5a,b).

The predictor showed strong discrimination. Mean AUROC was 0.974 across the four AllelePerturb datasets and 0.965 across all seven datasets, with near-perfect discrimination for JAK1 (0.999) and Replogle (0.981), and high discrimination even for TP53 (0.943; Fig. 5a). Thus, after a pilot assay, split-half effect-size estimates provide a reliable triage signal for identifying variants that lie within an evaluable window before full model benchmarking. Absolute calibration, however, remained dataset-dependent (Fig. 5b). The R-squared between predicted and observed un-rankable fractions was only 0.11, with a mean absolute error of 26 percentage points; GATA1 was the largest deviation (predicted 23%, observed 96%). The predictor is therefore best interpreted as a triage tool for prioritizing rankable perturbations, not as a universal calculator of absolute un-rankable rates across datasets.

The same analysis also reframes experimental depth as an effect-size-conditioned design variable (Fig. 5c). A strong-effect gene such as JAK1 crossed the detection floor from as few as 50 cells per half, whereas TP53 and KRAS remained near the noise floor despite median raw depths of 929 and 1,000 cells per variant. The relevant design target is therefore not a fixed cells-per-variant threshold, but the depth required for a given effect-size, inter-variant-difference and sampling-noise structure. In this sense, the bottleneck shifts from raw cell count to the effect-size-to-noise window that the experiment must clear.

As a practical check, we revisited the external gene-level atlases in an experimental-design context (Fig. 5d). At native depth, 55% of perturbations were un-rankable in Replogle, compared with 15% in Adamson and 3% in Norman. When normalized to the same shallow depth ($n = 50$ per half), the un-rankable fractions shifted toward one another: Replogle remained high at 55%, while Adamson increased to 39% and Norman to 11%. This pattern indicates that depth normalization can substantially change which perturbations are evaluable, while dataset-specific effect-size structure remains important. Together, these findings motivate a power-aware workflow for variant-level perturbation evaluation (Fig. 5e): estimate effect size and the split-half floor in a pilot assay; triage variants by predicted rankability; size the experiment to clear the effect-size-to-noise floor; and benchmark models only on perturbations whose ground truth is sufficiently measurable. This converts variant-level perturbation evaluation from uniform model ranking into a power-aware, stratified evaluation design.

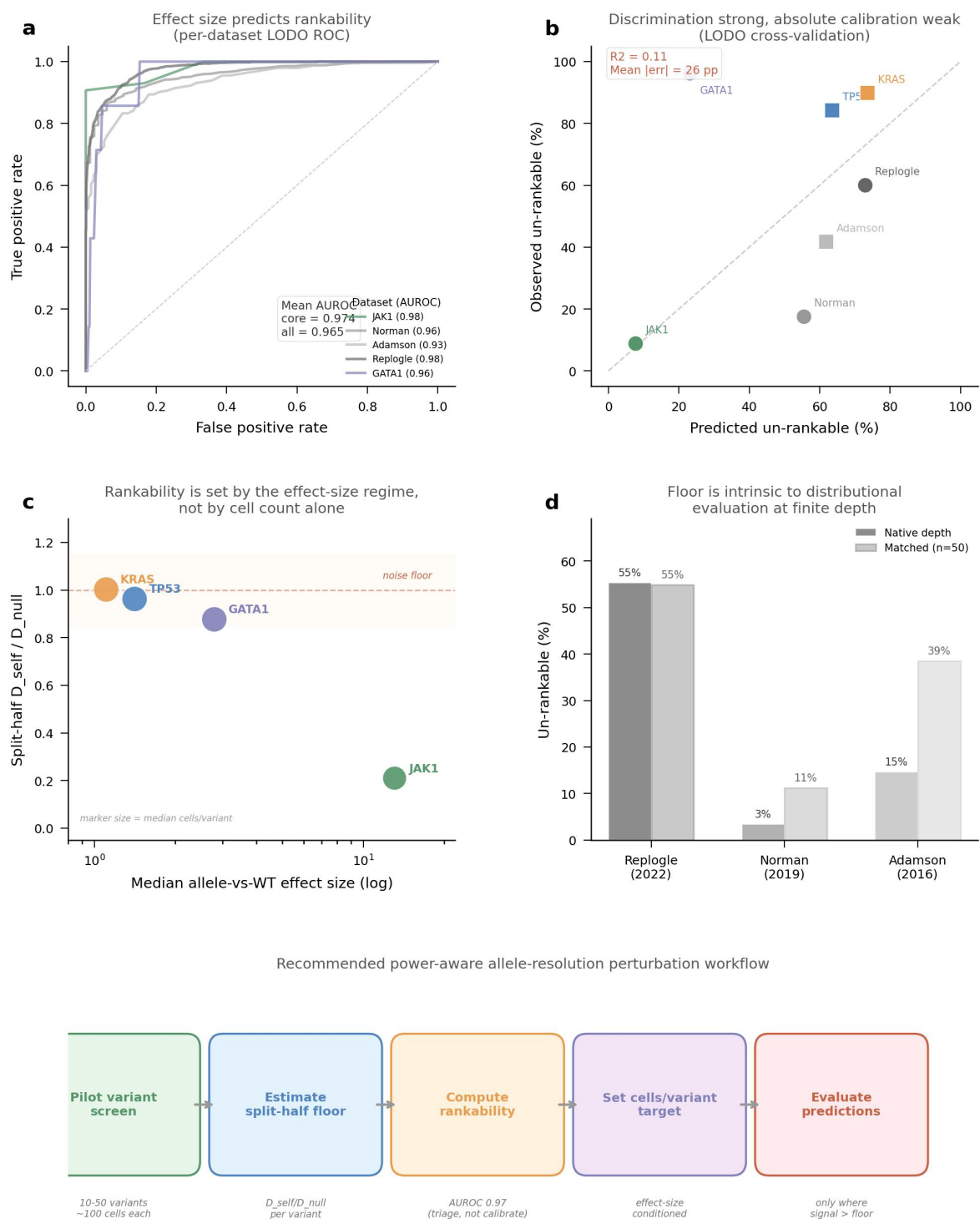


Figure 5: A power-aware reporting protocol for allele-resolution perturbation prediction. **a**, Effect size predicts rankability under leave-one-dataset-out cross-validation (mean AUROC 0.974 core, 0.965 all). **b**, Discrimination is strong but absolute calibration is weak ($R^2 = 0.11$, mean $|\text{err}| = 26$ pp). **c**, Rankability is set by the effect-size regime, not by cell count alone. **d**, The floor is intrinsic to distributional evaluation at finite depth, shown by native versus depth-matched ($n=50$) comparison (prescriptive; compare Fig. 3d). **e**, Recommended power-aware allele-resolution workflow.

Discussion

AllelePerturb shows that protein-coding variant prediction is not simply a finer-grained version of gene-level perturbation prediction. At this resolution, two notions of success that are often coupled in gene-level benchmarks separate: models can recover the shared transcriptional direction of a perturbation while failing to discriminate the responses of individual alleles. This direction-ranking dissociation reframes held-out variant prediction as a measurement-power problem. The central question is not only which model best predicts a perturbation, but whether the allele-specific component of the ground truth is measurable enough to support model comparison.

This finding has implications for both benchmark design and variant biology. Recent studies have shown that perturbation-prediction methods often differ little from simple baselines and that rankings can be sensitive to evaluation choices^{14,15}. AllelePerturb provides a complementary metrological explanation: when many perturbations lie below the resolution of distributional distance metrics, model rankings may be unstable regardless of model complexity. Biologically, direction recovery indicates that variants within a gene can share a program-level response, whereas allele discrimination requires resolving the finer signal that distinguishes variants such as TP53 R175H from R273C. Our results do not imply that allele-specific transcriptional effects are absent. Prior single-cell variant-impact phenotyping and the JAK1 positive control in AllelePerturb show that such effects can be detectable when the effect-size-to-noise window is sufficiently wide.

The practical consequence is a change in how variant-level perturbation benchmarks should be built. Rather than evaluating all variants uniformly, AllelePerturb-Eval suggests a power-aware workflow: estimate effect size and split-half noise in a pilot assay, triage perturbations by rankability, size the experiment to clear the effect-size-to-noise floor and benchmark models only on perturbations whose ground truth is sufficiently measurable. The LODO rankability predictor implements this idea as a triage tool, not as a universal calculator of absolute cell-number requirements. Several limitations define the current scope: AllelePerturb covers four genes and is enriched for missense variants; θ is a deliberately simple biophysical representation; split-half rankability has limited power at very low cell counts; and our audit of external perturbation models reflects public interfaces rather than the theoretical limits of those model families.

Crossing the detection limit will require progress in both experimental and computational design. Experimentally, larger saturation-variant and base-editing screens with adequate cells per variant, balanced controls and matched replicates will be needed to bring allele-specific ranking into an evaluable regime. Computationally, perturbation models must expose explicit protein-coding variant conditioning rather than treating variants as gene-level perturbation labels. AllelePerturb provides the benchmark, diagnostics and power-aware evaluation framework for this transition, defining when protein-coding variant prediction in single-cell transcriptomics is measurable enough to be meaningfully modeled.

Methods

Benchmark construction

Data sources. AllelePerturb-Bench integrates single-cell perturbation data for four protein-coding genes assayed across three experimental technologies. TP53 (98 variants) and KRAS (93 variants) were obtained from a Perturb-seq screen of coding variants in A549 lung adenocarcinoma cells (Gene Expression Omnibus GSE161824)¹². GATA1 (255 variants) was obtained from a base-editing screen in human hematopoietic stem and progenitor cells, using the processed expression matrix and predefined holdout splits distributed with PerturbNet¹³ (Hugging Face repository `cyclopeta/PerturbNet_reproduce`). JAK1 (26 variants) was obtained from a single-cell single-nucleotide-variant sequencing (scSNV-seq) experiment in HT-29 colorectal cancer cells stimulated

with interferon-gamma (European Nucleotide Archive PRJEB48915; processed objects from Zenodo record 10418435)¹⁷. Cell counts include both variant-assigned cells and the dataset-specific wild-type or control cells retained after preprocessing; the four datasets together comprise 472 variants and 321,043 cells.

Preprocessing. For TP53 and KRAS, processed count matrices, cell barcodes and variant-to-cell assignment tables were loaded from GEO, and cells were assigned to variants using the published assignment files. For GATA1, we used the PerturbNet-distributed AnnData object, which provides log-normalized expression over 2,477 highly variable genes together with five predefined hold-out partitions. For JAK1, the SingleCellExperiment object was loaded in R (Bioconductor); cells were filtered to homozygous genotyped calls with unambiguous coding consequences (missense, splice-site, stop-gained or synonymous), the 2,000 most highly variable genes were selected with `scrann::modelGeneVar` followed by `getTopHVGs`, and log-normalized counts were exported. Each dataset was processed in its own gene space, because the three cell types do not share a common highly-variable-gene set; the object shared across datasets is the variant feature representation, not the gene space. Within each dataset, every gene was standardized to zero mean and unit variance across all cells, and wild-type cells served as the common reference.

Variant feature vector. Each variant v was represented by protein-level features rather than by a variant identifier, so that held-out variants unseen in training can still be predicted. The primary representation is a six-dimensional biophysical vector

$$\theta_v = (\Delta H_v, \Delta V_v, \Delta Q_v, f_v, s_v, h_v),$$

where the first three components are the mutant-minus-wild-type differences in Kyte-Doolittle hydrophobicity, residue molecular weight and formal charge at pH 7.0. Writing $p(\cdot)$ for a residue property and $a_v^{\text{mut}}, a_v^{\text{wt}}$ for the mutant and wild-type amino acids of variant v , each biophysical difference is $\Delta p_v = p(a_v^{\text{mut}}) - p(a_v^{\text{wt}})$, standardized across all 472 variants by $z(\Delta p_v) = (\Delta p_v - \mu_{\Delta p}) / \sigma_{\Delta p}$. The remaining three components are structural indicators. The fold-core term f_v was set to 1.0 for residues within an annotated structural core (TP53 DNA-binding domain, residues 94 to 293; KRAS G-domain, residues 1 to 169; GATA1 zinc fingers ZF1 residues 204 to 228 and ZF2 residues 258 to 282; JAK1 kinase domain, residues 866 to 1154), to 0.5 for annotated interaction regions, and to 0.0 elsewhere. The functional-switch term s_v was set to 1.0 for catalytic or metal-coordinating residues (for example KRAS G12, G13 and Q61, and the GATA1 zinc-finger cysteines). The hotspot or pathogenic marker h_v was defined exclusively from external prior knowledge, with no reference to the single-cell outcome: canonical COSMIC and IARC missense hotspot codons for TP53 (175, 176, 179, 220, 238, 245, 248, 249, 273, 282, 285) and KRAS (12, 13, 59, 61, 117, 146); zinc-coordinating cysteines for GATA1 (204, 207, 225, 228, 258, 261, 279, 282) together with ClinVar pathogenic or likely-pathogenic annotations; and functional-module membership (JH2 pseudokinase, residues 583 to 855; JH1 kinase, residues 875 to 1153) for JAK1. An earlier version of h_v used the top quintile of measured effect size, which leaks the evaluation outcome into the feature; we removed it and re-ran the entire evaluation grid. Ranking performance was unchanged (θ -only PDS shifted by at most 0.02 and every method remained at or below chance), confirming that the central result does not depend on any outcome-derived feature.

Protein-language-model embeddings. As an alternative representation, we extracted per-variant embeddings from ESM-1v (`esm1v_t33_650M_UR90S_1`)¹⁸. Writing $\mathbf{H}_v \in \mathbb{R}^{L_v \times 1280}$ for the final-layer hidden states of the full-length mutant sequence of length L_v , the embedding is the mean over residue positions,

$$e_v = \frac{1}{L_v} \sum_{\ell=1}^{L_v} \mathbf{H}_v[\ell, :] \in \mathbb{R}^{1280},$$

used directly, without dimensionality reduction, as the ESM feature space.

Evaluation protocol (AllelePerturb-Eval)

Pseudobulk perturbation profiles. Let C_v and C_{WT} denote the cells of variant v and of wild type, and $x_i \in \mathbb{R}^G$ the standardized expression of cell i over G genes. The perturbation profile of variant v is the pseudobulk difference

$$\delta_v = \frac{1}{|C_v|} \sum_{i \in C_v} x_i - \frac{1}{|C_{WT}|} \sum_{j \in C_{WT}} x_j.$$

To bound the influence of the most deeply sampled variants, up to 300 cells were sampled without replacement per variant and per wild-type reference; variants with fewer than five cells were excluded. Every predictor maps a variant feature vector to an estimate $\hat{\delta}_v$ of this profile, and all metrics compare $\hat{\delta}_v$ to the measured δ_v .

AllelePerturb-Eval separates prediction quality into three families of metrics. **Direction recovery** asks whether a predicted profile points the right way in expression space. **Allele discrimination** asks whether a predicted profile is specific enough to identify its own variant among the other held-out variants. **Differential-expression fidelity** asks whether the predicted and measured profiles agree on which genes change and by how much. The ten metrics are defined as follows.

Perturbation discrimination score (PDS). PDS is the primary metric and directly tests allele discrimination: is a predicted profile closer to its own measured profile than to those of the other held-out variants? This nearest-match discrimination logic follows the perturbation-discrimination principle used in recent virtual-cell evaluation efforts¹⁶, adapted here to protein-coding variants and computed with tie-aware ranking. Let T be the set of $n_{\text{test}} = |T|$ held-out variants and $d(\cdot, \cdot)$ a distance. For variant v , the rank of the correct match among all candidates is

$$r_v = |\{w \in T : d(\hat{\delta}_v, \delta_w) < d(\hat{\delta}_v, \delta_v)\}| + 1,$$

with tied distances assigned the average rank, and the score is normalized so that the nearest match scores 1 and random matching scores 0.5 in expectation,

$$\text{PDS}(v) = 1 - \frac{r_v - 1}{n_{\text{test}} - 1}.$$

PDS was computed under three distances,

$$d_{\cos}(a, b) = 1 - \frac{a^\top b}{\|a\| \|b\|}, \quad d_1(a, b) = \|a - b\|_1, \quad d_2(a, b) = \|a - b\|_2,$$

giving PDS_cos (primary), PDS_L1 and PDS_L2. The WT-null reference predicts an identical zero profile for every variant, so all of its distances tie; average-rank assignment places it at exactly 0.5 under all three distances (the cosine distance to a zero vector is undefined and was treated as a uniform tie). Because the empirical chance level under small, gene-stratified test sets with ties can depart slightly from the analytic 0.5, we additionally estimated it by label permutation (see Statistics).

Direction-recovery metrics. Averaging the Pearson correlation over held-out variants gives

$$\text{Pearson-}\delta = \frac{1}{n_{\text{test}}} \sum_{v \in T} \text{corr}(\hat{\delta}_v, \delta_v),$$

and δ -cosine is the analogous average of $\hat{\delta}_v^\top \delta_v / (\|\hat{\delta}_v\| \|\delta_v\|)$. Pearson- δ (top-20) restricts the correlation to the 20 genes with the highest variance across the training-set profiles, testing whether agreement is concentrated in a small number of high-variance genes.

Differential-expression metrics. For each variant, differentially expressed genes were defined by a per-gene Welch t -test of variant against wild-type cells (up to 300 cells per group). Let $\mathcal{D}_v^{\text{true}}$ be the $k = 50$ genes with the largest absolute t -statistic and $\mathcal{D}_v^{\text{pred}}$ the k genes with the largest absolute predicted change. DE overlap is

$$\text{DE overlap}(v) = \frac{|\mathcal{D}_v^{\text{pred}} \cap \mathcal{D}_v^{\text{true}}|}{k}.$$

DE-LFC Spearman is the Spearman correlation between measured log-fold-changes and predicted changes over $\mathcal{D}_v^{\text{true}}$, and direction agreement is the fraction of genes in $\mathcal{D}_v^{\text{true}}$ for which $\text{sign}(\hat{\delta}_{v,g}) = \text{sign}(\delta_{v,g})$.

Reconstruction metric. Mean absolute error is $\text{MAE}(v) = G^{-1} \sum_g |\hat{\delta}_{v,g} - \delta_{v,g}|$, averaged over held-out variants.

Generalization splits

Six split strategies probe distinct axes of generalization. In each, models are trained on the training variants of a gene and evaluated on its held-out variants.

1. Random: 35% of variants per gene held out uniformly at random, using the predefined partition where one is available (GATA1).
2. Positional (out-of-distribution position): variants in the C-terminal half of each protein held out.
3. Mechanistic (out-of-distribution mechanism): variants at hotspot or functional-switch residues held out.
4. Cross-gene: all GATA1 variants held out with training on the remaining genes. This split is reported separately and excluded from the main grid, because the datasets do not share a gene space and cross-gene transfer is therefore not defined for a shared decoder.
5. Low-depth: variants with fewer than 200 cells held out, isolating the shallow-sampling regime.
6. Compatibility: held-out sets matched to the PerturbNet evaluation protocol for GATA1, enabling comparison with prior work.

Prediction methods

We evaluated 20 predictors: 18 feature-model combinations formed by crossing six regression heads with three feature spaces, plus two references. Each predictor learns a map $g_\phi : \mathbb{R}^p \rightarrow \mathbb{R}^G$ from a p -dimensional variant feature vector to a G -dimensional pseudobulk profile, fit by minimizing squared error on the training variants, $\min_\phi \sum_{v \in \text{train}} \|g_\phi(\text{feat}_v) - \delta_v\|_2^2$ (with the head-specific regularizer below), and scored by the identical protocol above.

Regression heads. Ridge regression (L2 penalty, $\alpha = 1.0$); Lasso regression (L1 penalty, $\alpha = 0.01$, up to 2,000 iterations); random forest (50 trees, maximum depth 6); gradient-boosted trees (50 estimators, maximum depth 3, wrapped for multi-output regression); k -nearest-neighbors regression ($k = 3$); and a multilayer perceptron (two hidden layers of 128 and 64 units, ReLU activation,

up to 500 iterations). Linear and neighbor heads regress directly onto the full expression profile. Because tree ensembles scale poorly to thousands of output genes, for output spaces larger than 200 genes the random forest and gradient-boosted heads were fit on a q -component PCA of the training profiles and predictions were mapped back to gene space by the inverse transform, with $q = \min(50, G, n_{\text{train}} - 1)$. All heads used a fixed random seed.

Feature spaces. θ (six-dimensional biophysical vector); ESM (1,280-dimensional ESM-1v embedding); and ESM+ θ (their 1,286-dimensional concatenation).

References. The WT-null reference predicts $\hat{\delta}_v = 0$ for every variant, quantifying the score attainable with no perturbation signal. The Gene-mean reference predicts the mean training-variant profile of the held-out variant's own gene, $\hat{\delta}_v = |\text{train}(g_v)|^{-1} \sum_{w \in \text{train}(g_v)} \delta_w$; because it uses the gene label g_v , it has access to information a variant-blind predictor does not, and we report it as an informative upper reference on how much of the response is explained by gene identity alone rather than as a competing method. Gene-mean is defined only for genes with at least one training variant in a split and is therefore omitted under the cross-gene split.

External methods. We attempted to include six published perturbation-prediction methods. CPA² exposes a scalar dosage interface and returned undefined losses when supplied with multi-dimensional θ vectors. CellFlow and scDFM provide no inference path for variants absent from training. GEARS⁴ is keyed to a gene-level knowledge graph and is allele-blind by construction. STATE required per-dataset retraining and exceeded available GPU memory on GATA1. Biolord¹⁹ was the only external method that ran to completion, treating θ components as ordered attributes; its PDS fell between 0.47 and 0.51, within the chance band, consistent with the 18 in-house feature-model combinations.

Split-half rankability analysis

To separate model limitation from measurement limitation, we asked whether variant responses are stably measurable at the available sampling depth. The energy distance between two cell sets $X = \{x_i\}_{i=1}^n$ and $Y = \{y_j\}_{j=1}^m$ is

$$E(X, Y) = \frac{2}{nm} \sum_{i=1}^n \sum_{j=1}^m \|x_i - y_j\| - \frac{1}{n^2} \sum_{i,i'} \|x_i - x_{i'}\| - \frac{1}{m^2} \sum_{j,j'} \|y_j - y_{j'}\|,$$

with Euclidean norms. For each variant with at least 50 cells, cells were partitioned into two random halves A and B , and a size-matched wild-type control sample C was drawn. We computed the within-variant replicate distance $D_{\text{self}} = E(A, B)$ and the variant-to-wild-type distance $D_{\text{null}} = E(A, C)$. Distances were evaluated over a grid of subsample sizes per half (30, 50, 100, 150, 250 and 365 cells for variant-level datasets) and in three representation spaces (a 50-component PCA, the gene space and a pathway-score space), with 50 random seeds per configuration; the canonical values use the PCA space at native maximum depth. Each seed distribution was summarized by its median and its 2.5 to 97.5 percentile interval.

A variant was classified as rankable when its signal exceeded the sampling width of the replicate distance,

$$S = \text{med}(D_{\text{null}}) - \text{med}(D_{\text{self}}) > W = Q_{97.5}(D_{\text{self}}) - Q_{2.5}(D_{\text{self}}),$$

and un-rankable otherwise. This signal-versus-noise criterion is the primary definition; we confirmed that the conclusions are qualitatively unchanged under four alternatives (lower confidence bound of S above zero; $S > 1.5W$; $S > 2W$; and a ratio-based threshold), which shift absolute fractions but preserve the ordering across datasets (Extended Data Fig. 1). The same analysis

was applied to the four AllelePerturb genes and to three widely used gene-level Perturb-seq atlases, Replogle 2022, Norman 2019 and Adamson 2016 (K562 cells), obtained from the scPerturb data portal⁷, using an identical native-depth, all-perturbation, direct split-half definition throughout.

Rankability predictor

To test whether rankability can be anticipated before a full model comparison, we trained a logistic-regression classifier to predict per-perturbation rankability under leave-one-dataset-out cross-validation across all seven datasets. The four features, all estimable from a pilot experiment before modeling, were $\log_{10}$ of cells per perturbation, the pseudobulk effect size $\|\delta_v\|_2$, the PCA explained-variance ratio and the gene-space dimensionality. Performance was measured by the area under the receiver-operating-characteristic curve (discrimination) and by comparing predicted with observed un-rankable fractions per held-out dataset (calibration).

Power analysis

To relate detection to sampling depth, cells were subsampled at 50, 100, 150 and 300 per variant and the split-half rankability test was recomputed at each depth. The fraction of variants classified as detectable was plotted against subsample size for each gene, and a detection threshold (the depth at 50% detection) was estimated by sigmoid fitting where the data supported it.

Statistics

Confidence intervals for PDS and Pearson- δ were obtained by nonparametric bootstrap over held-out variants (2,000 resamples, percentile method, 95% level). A method was called significantly below chance when the upper bound of its PDS interval fell below 0.50, and significantly positive on direction when the lower bound of its Pearson- δ interval exceeded 0. The per-gene $D_{\text{self}}/D_{\text{null}}$ ratio (Fig. 3b) and the external-atlas rankability fractions (Fig. 3d, 3e) were bootstrapped over variants and over perturbations, respectively. To confirm that the analytic PDS chance level of 0.50 holds under the actual tie structure and small test sets, we ran a within-split label-permutation control: for each method-split-gene combination the full prediction-to-truth distance matrix was precomputed and the prediction-to-target correspondence was permuted 1,000 times, recomputing PDS with the same tie-aware ranking each time. The empirical null mean was 0.500 (range 0.495 to 0.504 across 340 combinations), and only 3.2% of combinations exceeded the permutation 97.5 percentile, below the 5% expected by chance. All bootstraps and permutations used fixed random seeds. We report intervals to characterize uncertainty rather than as per-method hypothesis tests: the conclusions rest on the direction and consistency of the effect across all 19 non-reference predictors, which move together, and so do not depend on any single comparison or on a multiplicity correction.

Software and reproducibility

The evaluation grid was run in Python 3.11 with NumPy 2.4.6, pandas 2.3.3, scikit-learn 1.7.2 and SciPy 1.14.1. Single-cell loading and preprocessing used scanpy 1.10 (Python) and scanr with SingleCellExperiment (R, Bioconductor). ESM-1v embeddings were extracted with fair-esm 2.0.0 and PyTorch 2.10. Figures were rendered with matplotlib 3.10. All code, processed benchmark tables and evaluation protocols are available at <https://github.com/Boom5426/AllelePerturb>.

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